

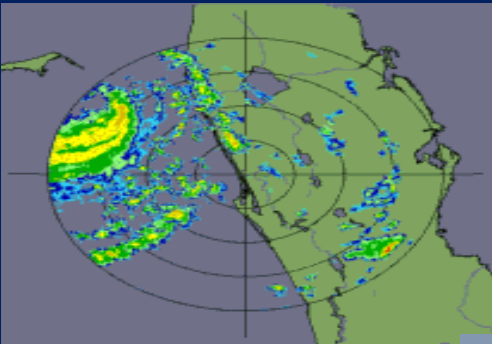


USPCAS-W
Mehran University

HYDROINFORMATICS

GIS Data Sources and Acquisition

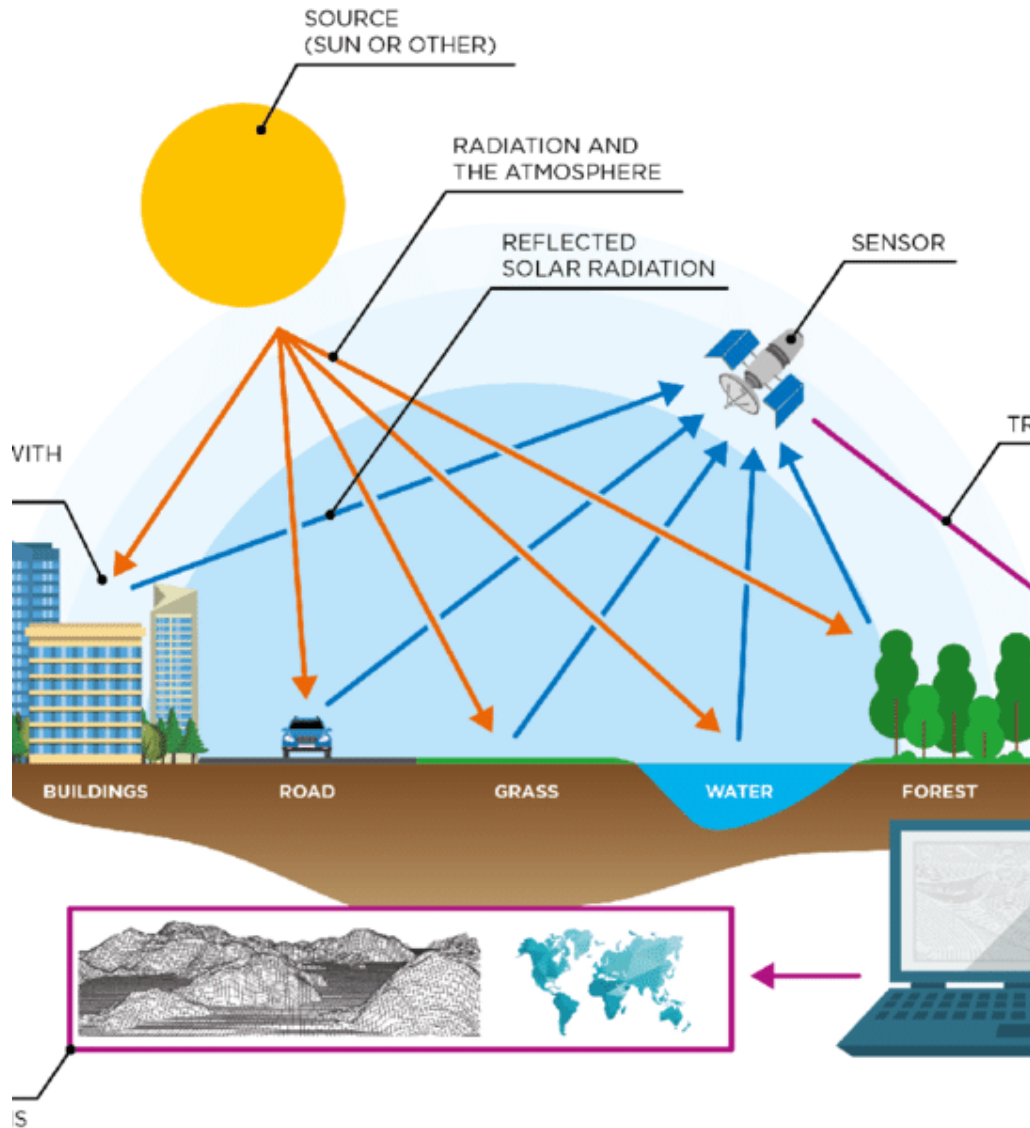
By: Dr. Arjumand Zaidi
Senior Research Fellow



Fall 2026
US Pakistan Center for Advanced Studies in **Water**
MUET, Jamshoro



Let's kickstart our brains on this topic.



- What are some common on-the-ground methods for data collection?
- What are the primary sources of satellite data for Earth observation?



Learning Objectives

1. To discuss data acquisition challenges.
2. To learn about free satellite data sources.

Data Collection Methods

- **Secondary Data Collection**

- existing data
- already published/available
- Someone else collected data

- **Primary Data Collection**

- from scratch
- You collected or assigned someone to collect

Field Measurements

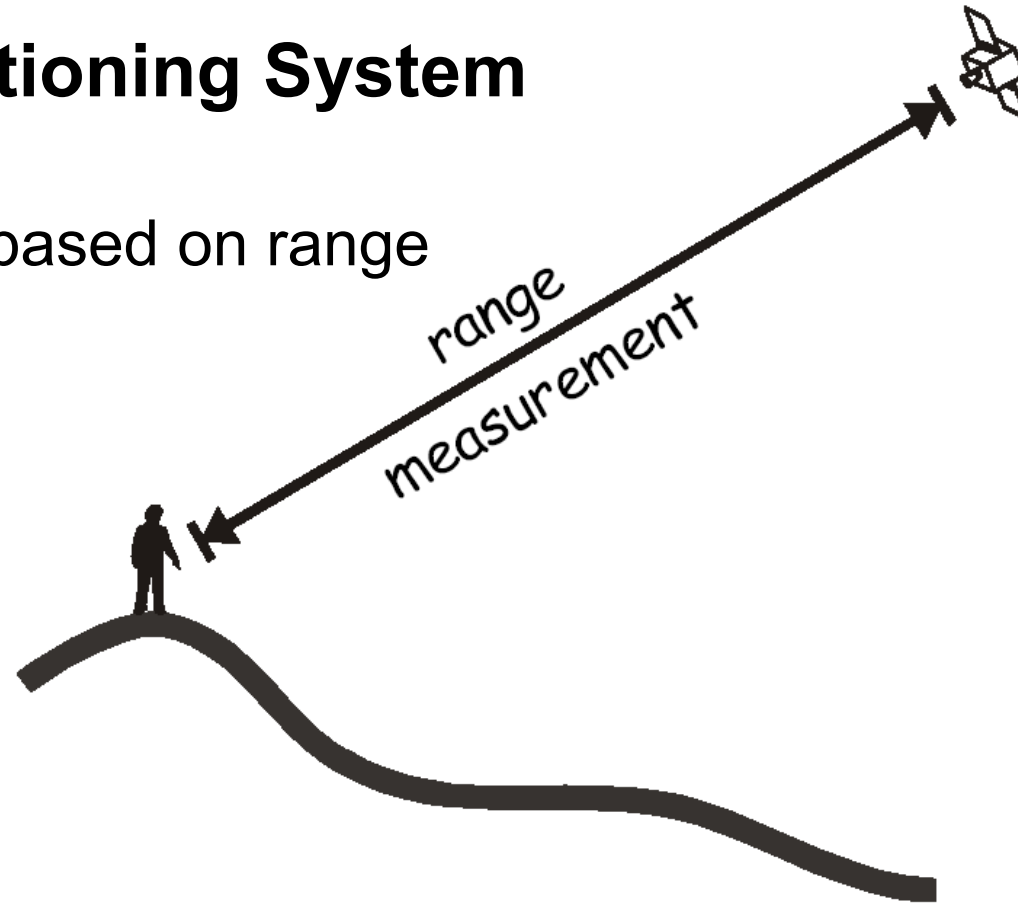


Coordinate Surveying



GPS: Global Positioning System

Position is estimated based on range measurements.



Range = speed of light x travel time

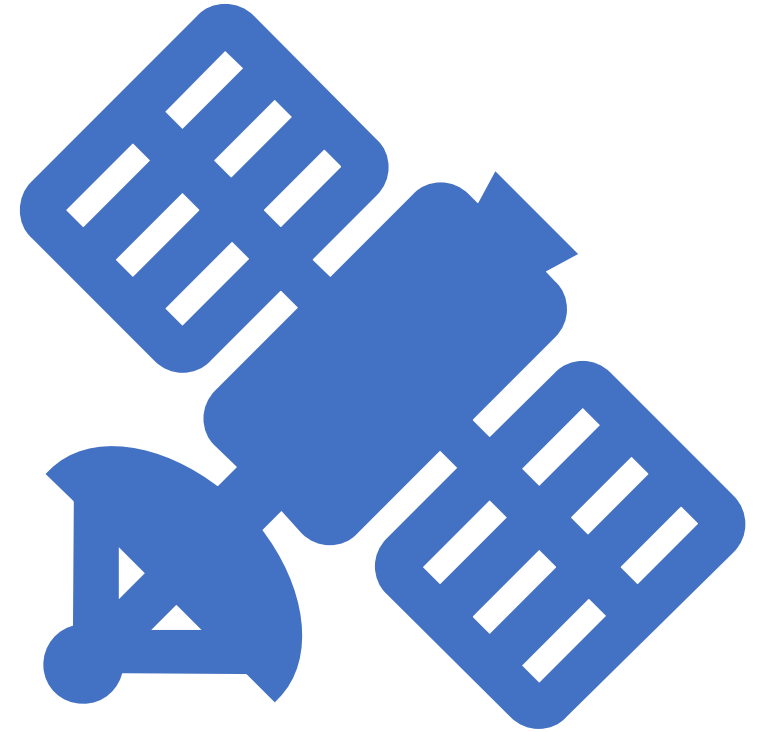
$$\text{Range} = c(t/2)$$

(c = 299,792,458 meters per second, t = round trip time of a signal)

Remote Sensing: measuring electromagnetic energy reflected or emitted from objects – airborne or satellite-based instruments



satellites vary by: orbit, altitude, resolution, revisit frequency, revisit variability (steering) capability, width of swath, image size, stereo capability, wavelengths collected, other sensors, customer delivery time, etc.



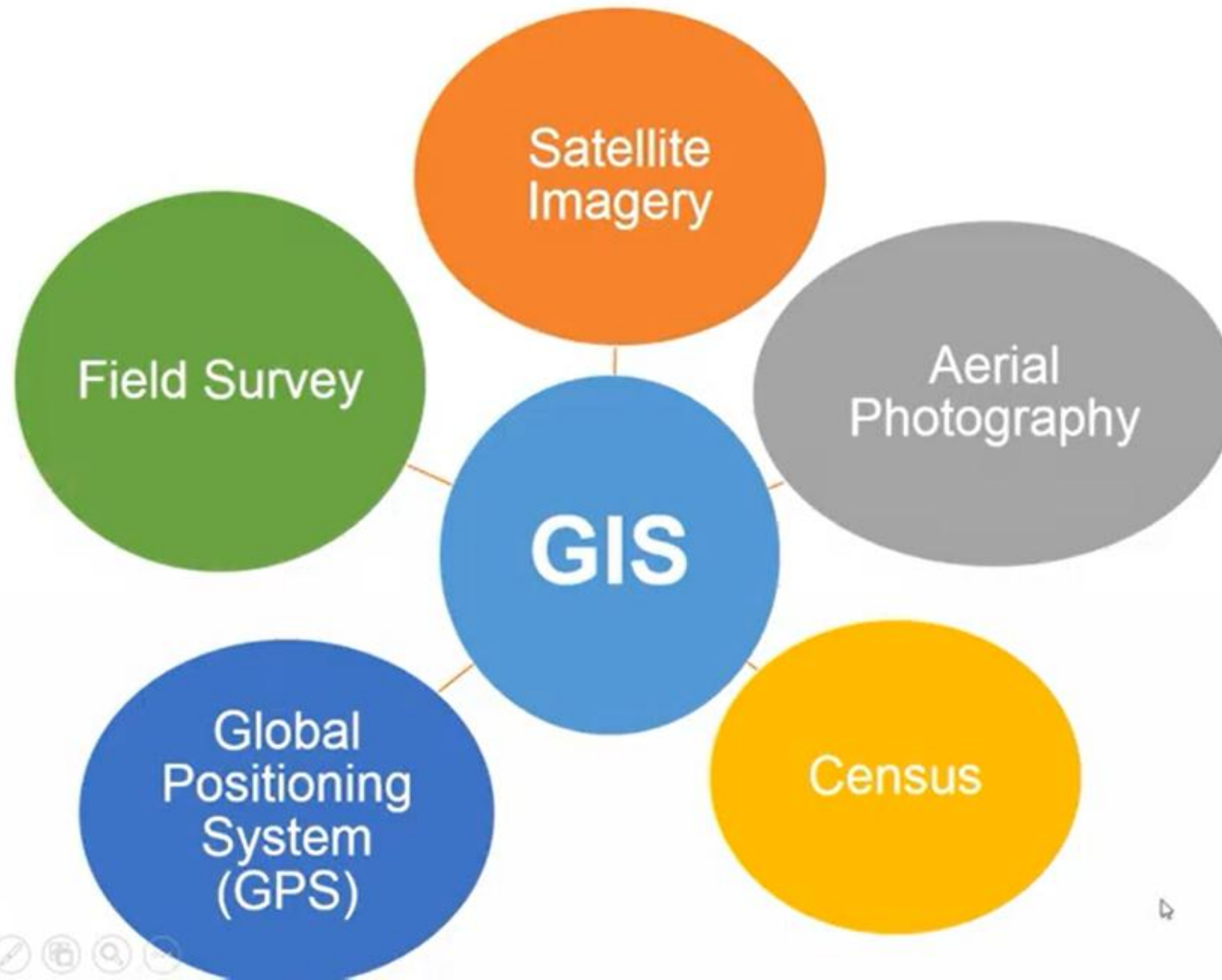
Data Collection Protocols

- Employ a reliable sampling method for representative data.
- Standardize procedures and employ quality control to enhance data reliability and consistency.
- Minimize bias in primary data collection.
- Obtain consent and safeguard participants' confidentiality and rights.

Data use protocols

- Credit secondary data sources through proper citation.
- Ensure data is obtained and used ethically.
 - Adhere to pertinent laws/regulations to ensure legal conformity, safeguard intellectual property rights, and uphold ethical standards.
- Assess the quality and reliability of data before utilization.

Sources of GIS Data



GIS Data Sources

Maps and Drawings

- Scanning or digitizing

Aerial/Satellite Images

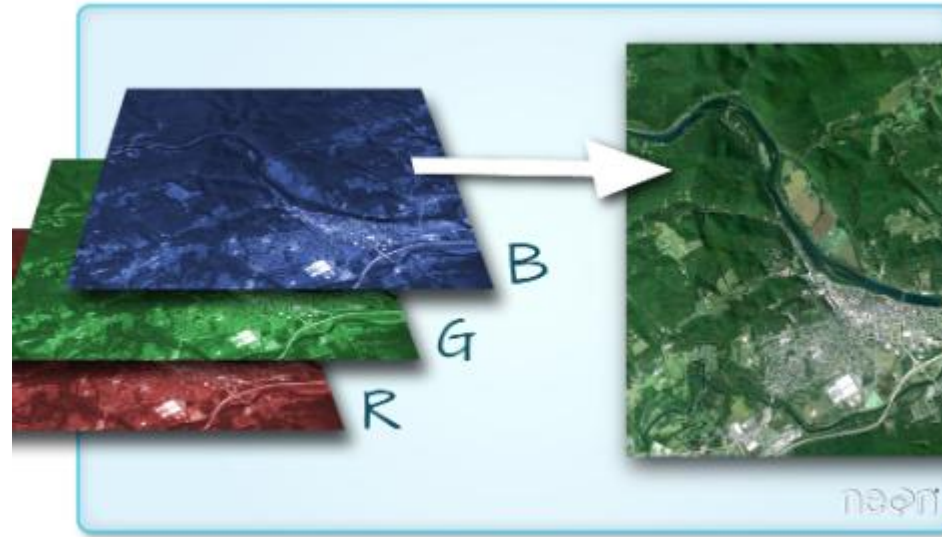
- Image interpretation for features extraction
- Digital terrain model) to create digital orthos

CAD Databases

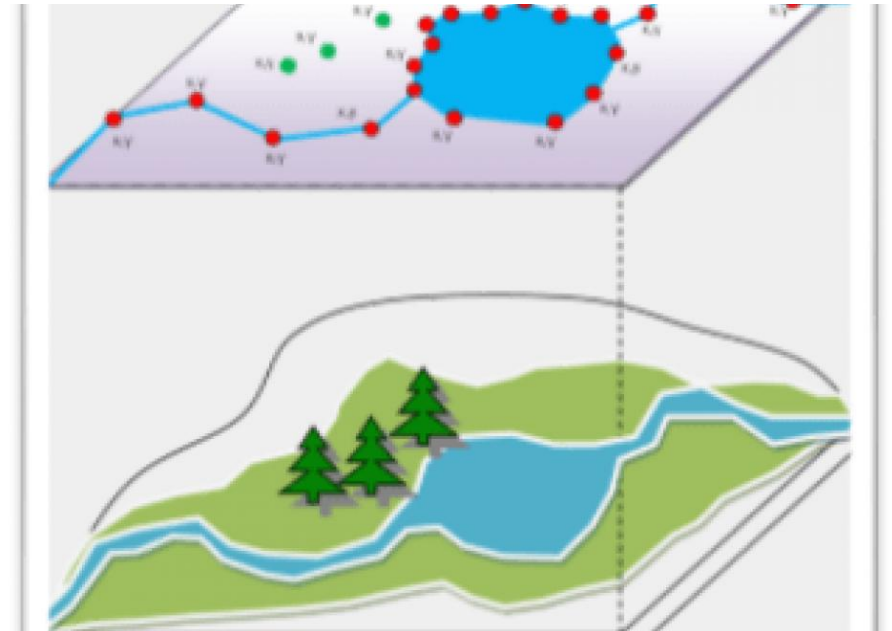
GIS Databases

Attribute Databases

Records and Documents



Sources of Vector/Raster Data



ArcGIS Open Data

Steer decisions and results with your data

Data is critical to decision making. Whether you are buying a house, wondering about road construction, or opening a new business, trustworthy information is essential. Use Esri's SaaS open data capabilities to provide your community with the authoritative data they need to make better decisions and accomplish their goals.

<https://www.esri.com/en-us/arcgis/products/arcgis-open-data>

Landcover Explorer

<https://livingatlas.arcgis.com/landcoverexplorer/>

The screenshot displays the Landcover Explorer web application interface. The main map shows a satellite view of the Middle East and South Asia, with labels for cities like Ashgabat, Mashhad, Kabul, Srinagar, Islamabad, Lahore, Meerut, New Delhi, Kathmandu, Jaipur, Agra, Lucknow, Kanpur, Patna, Varanasi, Hyderabad, Jodhpur, Multan, Quetta, Zahedan, Kerman, Shiraz, Basra, Ahvaz, Kuwait City, Manama, Doha, and Dubai. A yellow circle highlights the left sidebar menu, which includes 'Esri | Sentinel-2 Explorer', 'Imagery Explorer Apps' (with sub-items: Landsat Explorer, Sentinel-1 Explorer, Sentinel-2 Land Cover Explorer, and NLCD Land Cover Explorer), 'Imagery Utilities', and 'Spectral Sampler'. A popup window titled 'Sentinel-2 | Aug 17, 2026' displays the following data: NDMI: -0.151, NDVI: 0.072, MNDWI: -0.368, and coordinates x: 62.523, y: 36.396. The bottom of the interface features three main sections: 'EXPLORE' with buttons for SWIPE, ANIMATE, and ANALYZE; 'DYNAMIC VIEW' with a 'FIND A SCENE' button and a description of the dynamic scene rendering process; and 'INTERESTING PLACES' with a grid of 12 thumbnail images. The 'RENDERER' section on the right shows a grid of 9 thumbnail images for different rendering styles: Natural Color, Agriculture, Color IR, Short-wave IR, Urban, NDVI, NDMI, MNDWI, and Bathymetric.

Esri | Sentinel-2 Explorer

Imagery Explorer Apps

- Landsat Explorer
- Sentinel-1 Explorer
- Sentinel-2 Land Cover Explorer
- NLCD Land Cover Explorer

Imagery Utilities

- Spectral Sampler

Sentinel-2 | Aug 17, 2026

NDMI: -0.151
NDVI: 0.072
MNDWI: -0.368
x: 62.523 y: 36.396

PAKISTAN

NEPAL

powered by Esri | Sentinel-2 imagery courtesy of European Space Agency, European Commission, and Microsoft

1:10,317,028 | 1px: 2,730m

EXPLORE

DYNAMIC

FIND A SCENE

DYNAMIC VIEW

In the current map display, the most recent and most cloud free scenes from the Sentinel-2 archive are prioritized and dynamically fused into a single mosaicked image layer. As you explore, the map continues to dynamically fetch and render the best available scenes.

To select an individual scene for a specific date, try the [FIND A SCENE](#) mode.

INTERESTING PLACES

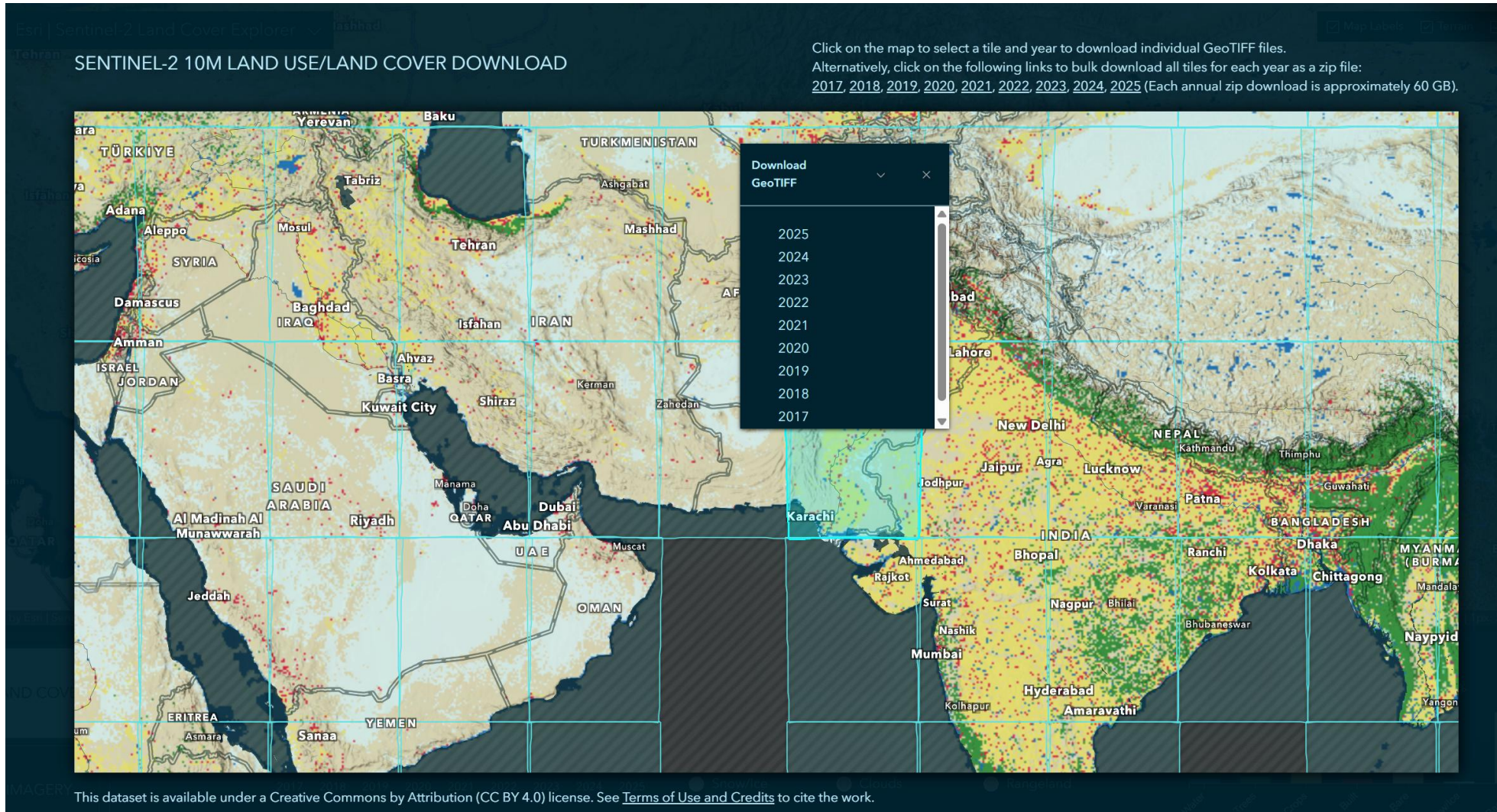
Renderer

Natural Color Agriculture Color IR

Short-wave IR Urban NDVI

NDMI MNDWI Bathymetric

ESRI LULC 2017 - 2025



<https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=70.32492%2C30.11343%2C5.88&mode=step&timeExtent=2017%2C2025&year=2025&downloadMode=true>

Natural Earth Data

Link: <https://www.naturalearthdata.com/>



The image shows the Natural Earth website interface. At the top, there is a dark blue header with a globe icon on the left, the text "Natural Earth" in the center, and a search bar on the right. Below the header is a navigation bar with links: Home, Features, Downloads, Blog, Issues, Corrections, and About. The main content area features a map of the United States with a red "New!" banner in the top left corner. The map is labeled "Railroads" and "Map Gallery". Below the map, there is a paragraph of text describing the dataset. At the bottom, there is a green box with text about the project's history and a green button labeled "Get the Data".

New!

Map Gallery

Railroads

Natural Earth is a public domain map dataset available at 1:10m, 1:50m, and 1:110 million scales. Featuring tightly integrated vector and raster data, with Natural Earth you can make a variety of visually pleasing, well-crafted maps with cartography or GIS software.

Natural Earth was built through a collaboration of many **volunteers** and is supported by **NACIS** (North American Cartographic Information Society), and is free for use in any type of project (see our **Terms of Use** page for more information).

Get the Data

USGS Earth Explorer

Link: <https://earthexplorer.usgs.gov/>

The screenshot displays the USGS Earth Explorer web application. At the top, the USGS logo is accompanied by the tagline "science for a changing world". Below this, the "EarthExplorer" title is on the left, and navigation links for "System Notification (1)", "Help", "Feedback", and "Login" are on the right. The main interface is divided into two primary sections. On the left, the "Search Criteria" tab is active, showing a "1. Enter Search Criteria" section with instructions on how to narrow search areas. It includes a "Geocoder" sub-tab and a dropdown menu for "Select a Geocoding Method" currently set to "Feature (GNIS)". Below this, there are search limits, tabs for "US Features" and "World Features", and input fields for "Feature Name", "State", and "Feature Type". On the right, the "Search Criteria Summary (Show)" tab is active, displaying a satellite map of the central United States. The map shows state boundaries and labels for cities including Aberdeen, Willmar, Watertown, Minneapolis, St. Paul, Rochester, Austin, Albert Lea, Sioux Falls, Pierre, and Mason City. A coordinate box in the top right of the map shows "(42° 49' 10\" N, 100° 47' 18\" W)". A "Clear Search Criteria" link is located at the top right of the map area. An "Activate Windows" watermark is visible in the bottom right corner of the map.

OPENSTREETMAP
AN OPEN SOURCE MAP PROJECT

OpenStreetMap | History | Export | GPS Traces | User Diaries | Communities | Copyright | Help | Donate | About

Search: Where is this?

Welcome to OpenStreetMap!

OpenStreetMap is a map of the world, created by people like you and free to use under an open license.

Hosting is supported by Fastly, OSMF corporate members, and other partners.

By using this website or other infrastructure provided by the OpenStreetMap Foundation, you agree to the Terms of Use.

[Learn More](#) [Start Mapping](#)

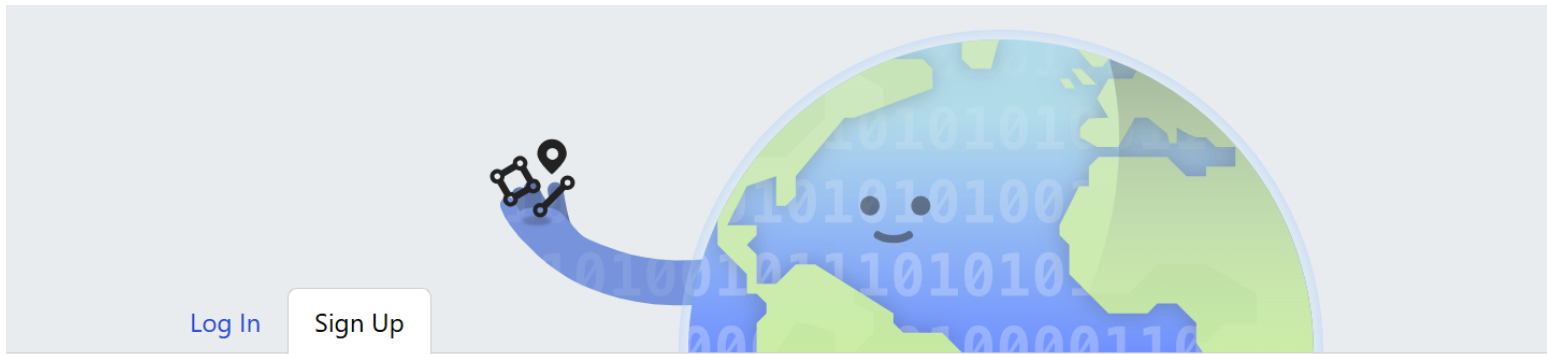
Map Layers

- Standard
- CycloSM
- Cycle Map
- Transport Map
- Tracestrack Topo
- Humanitarian
- Shortbread
- MapTiler OMT

Enable overlays for troubleshooting the map

- ☐ Map Notes
- ☐ Map Data
- ☐ Public GPS Traces

<https://www.openstreetmap.org/#map=6/30.67/69.36>



Free and editable. Unlike other maps, OpenStreetMap is completely created by people like you, and it's free for anyone to fix, update, download and use.

Sign up to get started contributing.

Email

Your address is not displayed publicly, see our [privacy policy](#) for more information.

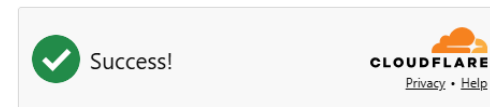
Display Name

Your publicly displayed username. You can change this later in the preferences.

Password

Confirm Password

Verify you're a human



By signing up, you agree to our [Terms of Use](#), [privacy policy](#) and [contributor terms](#).

Sign Up

Another Option for Downloading OSM is from GeoFabrik Downloads

<https://www.geofabrik.de/data/download.html>



Our Download Server

Our download server at <https://download.geofabrik.de> has excerpts and derived data from the [OpenStreetMap dataset](#) available for free download. Most of these files are updated every day – any change you upload to OpenStreetMap should be on our download server the next day.

Geographic Regions

Data on the download server is organised by region. The root directory contains files that have a whole continent's data in them, and for all continents except Antarctica there are subdirectories in which you find individual files for various countries. Some countries again have their own subdirectories with data for administrative subdivisions. For Germany, all "Bundesländer" are available as separate files, and for England we have all individual counties. For France, we have all (historic) Departements. The file for Germany's Baden-Württemberg region is, for example, at

<https://download.geofabrik.de/europe/germany/baden-wuerttemberg-latest.osm.pbf>

Please let us know if you are interested in other regional excerpts that we do not offer right now – we might be able to add them with little effort.

Raw data (OSM source data)

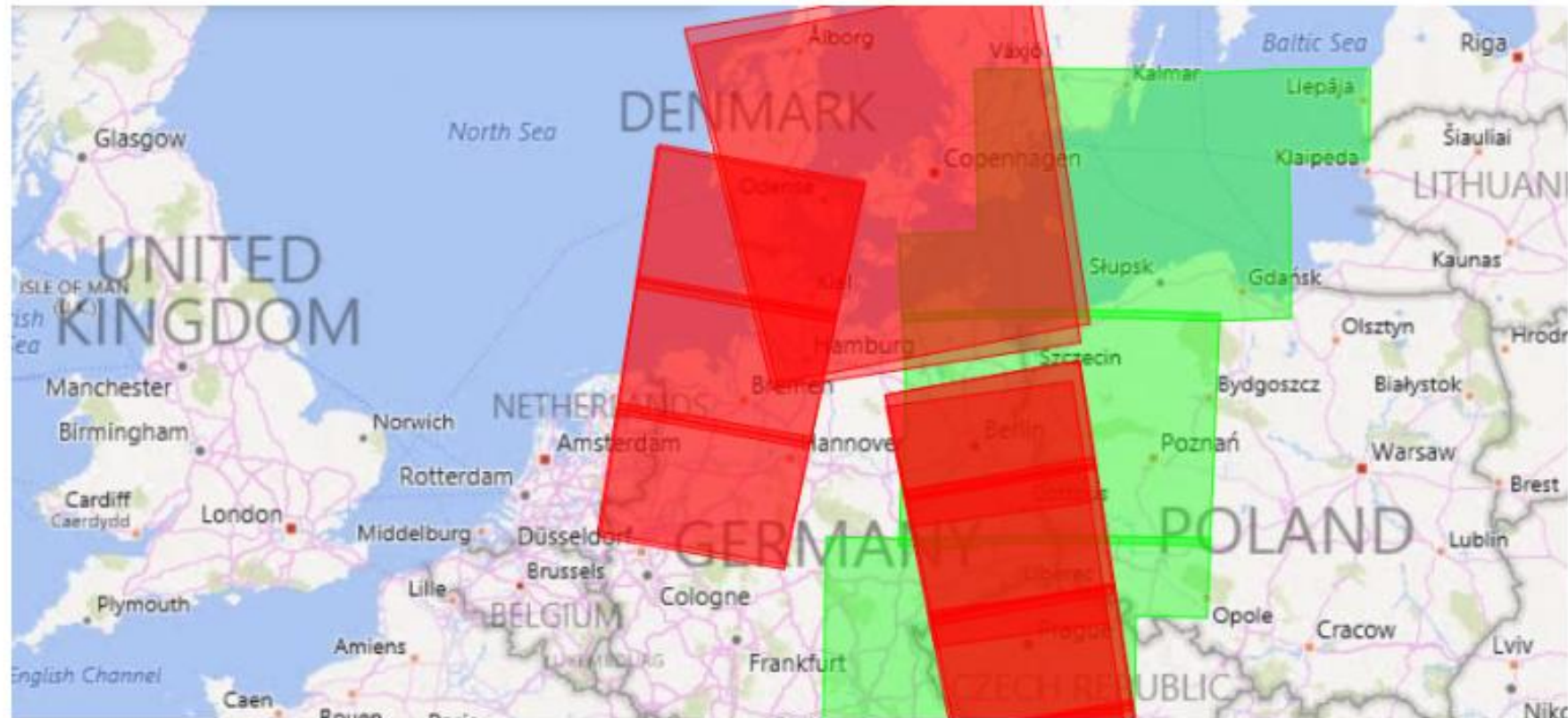
Files with the `.osm.pbf` suffix are OpenStreetMap raw data files in the [PBF format](#). They are the common format for OpenStreetMap raw data and need few memory for storage. Daily diff updates (`.osc.gz`, Gzip compressed OSM XML) are available on our download server as well. They contain the changes of one day. The PBF files can be updated with them.

The Bzip2 compressed XML files we used to have on the download server are now discontinued since almost all software supports `.osm.pbf`. If you still need bz2 files, you can [create them yourself](#), or order them commercially from us.

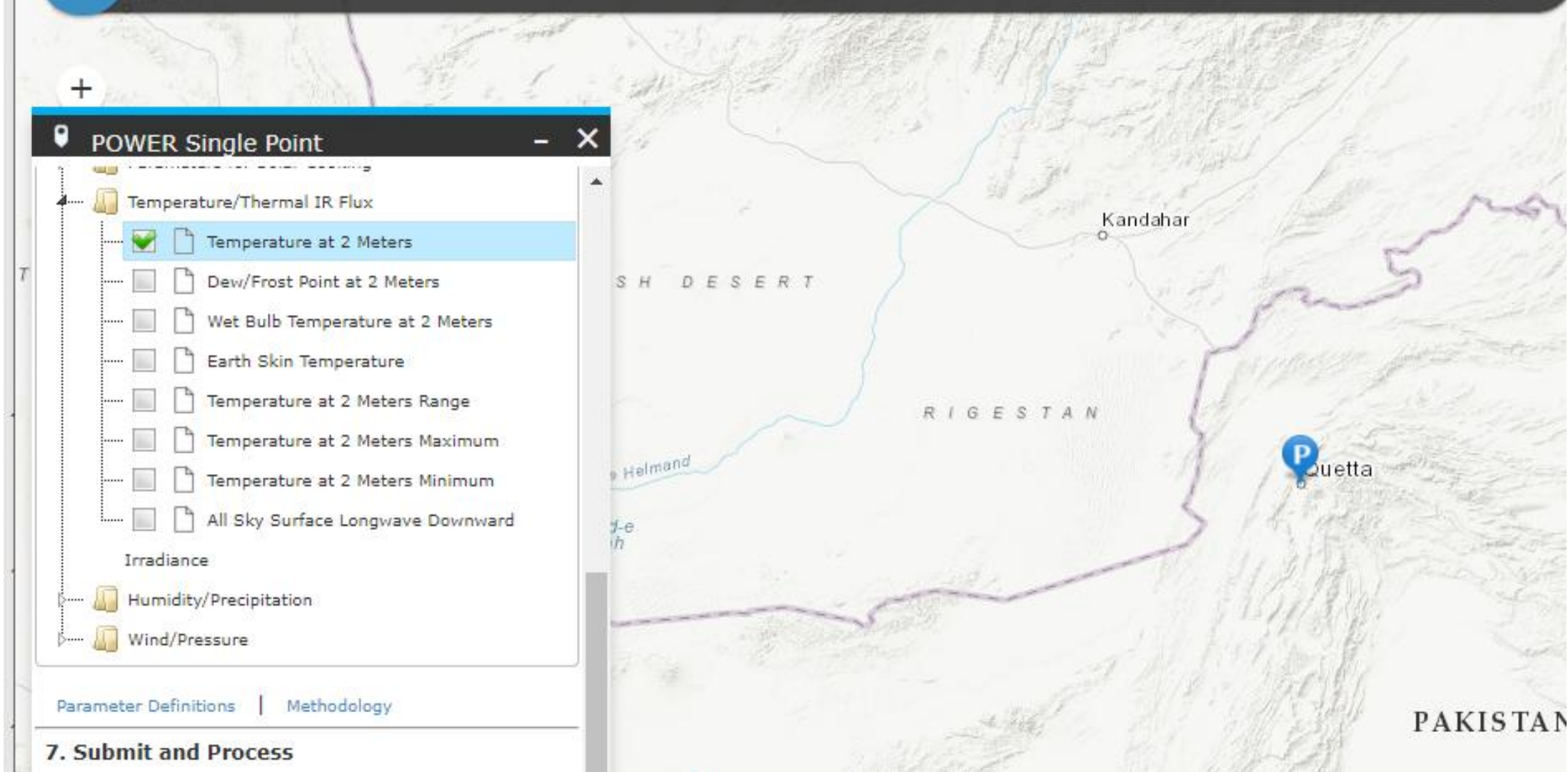
You need special software to process OSM raw data. It is usually command line and server software for Linux. Common tools to process OpenStreetMap raw data are:

How to Download Free Sentinel Satellite Data

By: GISGeography • Last Updated: October 1, 2023

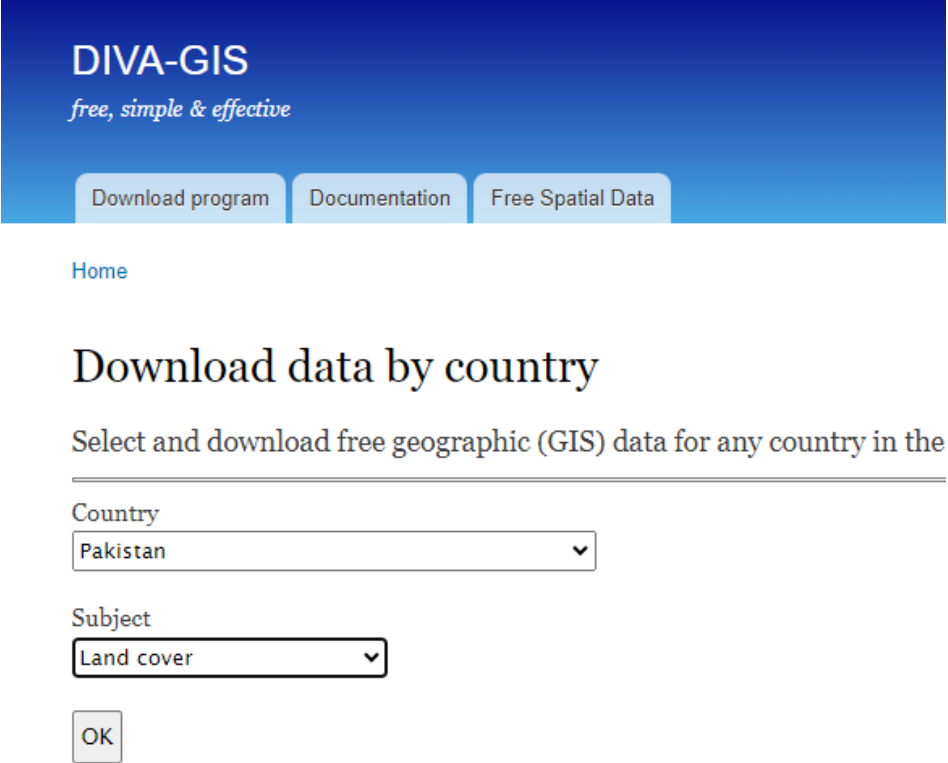


Download Sentinel-1 and Sentinel-2 Satellite Data



Weather Data

Link: <https://power.larc.nasa.gov/data-access-viewer/>



Arces

Subject	Description	Source	Format	Resolution
Administrative boundaries	Country outlines and administrative subdivisions for all countries. The level of subdivision varies between countries	GADM , version 1.0	Vector (area)	-
Rivers and water	Rivers, canals, and lakes. Separate files for line and area features	Digital Chart of the World	Vector (line and area)	-
Roads	Roads	Digital Chart of the World	Vector (line)	-
Railroads	Railroads	Digital Chart of the World	Vector (line)	-
Elevation	SRTM30 dataset. CGIAR-SRTM data aggregated to 30 seconds	CGIAR SRTM (3 seconds resolution)	Grid	30 s
Land cover	Land cover, original data resampled onto a 30 seconds grid	GLC2000	Grid	30 s
Population	Population density (old)	CIESIN , 2000. Global gridded population database	Grid	30 s
	Monthly climate data	WorldClim	Grid	30 s

Diva-GIS:

- Download Data by Country

<https://www.diva-gis.org/gdata>

Diva GIS

Download Data by Country

<https://diva-gis.org/data.html>

Data

Select and download free  geographic (GIS)  data for any country in the world
  GIS data download

Country

Pakistan

Subject

Administrative areas

Format

Download **PAK_adm.zip**

Sources

Subject	Description	Source	Format	Resolution
Administrative areas (boundaries)	Country outlines and administrative subdivisions for all countries. The level of subdivisions that is available varies between countries	GADM	Vector (area)	-
Inland water	Rivers, canals, and lakes. Seperate files for line and area features	Digital Chart of the World	Vector (line and area)	-
Roads	Roads	Digital Chart of the World	Vector (line)	-
Railroads	Railroads	Digital Chart of the World	Vector (line)	-
Elevation	SRTM30 dataset. CGIAR-SRTM data aggregated to 30 seconds	CGIAR SRTM (3 seconds resolution)	Grid	30 seconds
Land cover	Land cover, original data resampled onto a 30 seconds grid	GLC2000	Grid	30 seconds
Population	Population density (old)	CIESIN , 2000. Global gridded population database	Grid	30 seconds
Climate	Monthly climate data	WorldClim	Grid	30 seconds
Gazetteer	A gazetteer is a list of place names and their coordinates. The files you can download here are for use in DIVA for automatic georeferencing (to assign coordinates to places). The files should be placed in the \\gazet directory. This is very old now. There area better services for this.	DBF	-	

Formats

The files have been compressed and grouped in ZIP files. You can use programs such as 7-zip, PKZIP or Stuffit to decompress the files.

Vector data are stored as ESRI shapefiles Grid (raster) data are stored as DIVA gridfiles

FAO SOILS PORTAL

1 km resolution

<https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v20/en/>



Food and Agriculture Organization
of the United Nations

ENHANCED BY Google

English Français Español

FAO SOILS PORTAL

[Home](#) [Data Hub](#) [Assessment](#) [Biodiversity](#) [Management](#) [Degradation/Restoration](#) [SoiLEX](#) [Publications](#)

Soil properties

Soil classification

Sampling and laboratory techniques

Soil Maps and Databases

Global Map of Salt-affected Soils

Global Soil Organic Carbon Sequestration Potential Map (GSOSeq)

Global Soil Organic Carbon Map (GSOcmap)

FAO/UNESCO Soil Map of the World

Harmonized world soil database v1.2

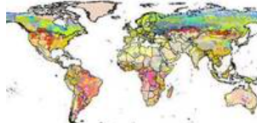
Harmonized world soil database v2.0

Other Global Soil Maps and Databases

Regional and National Soil Maps and Databases

Soil Profile Databases

Harmonized World Soil Database v2.0



The Harmonized World Soil Database version 2.0 (HWSD v2.0) is a comprehensive global soil inventory that offers detailed insights into soil properties, including the morphology, chemistry, and physical characteristics, with a focus on a 1 km resolution. It's a valuable tool for research in areas like agriculture, food security, and climate change impacts.

Developed in 2008 by the International Institute for Applied Systems Analysis (IIA) and the Food and Agriculture Organization of the United Nations (FAO), the HWSD was a collaborative project involving other notable organizations like the International Soil Reference and Information Centre (ISRIC), the European Soil Bureau Network (ESBN), and the Institute for Soil Sciences at the Chinese Academy of Sciences (CAS). The data was compiled and organized using a Geographic Information System (GIS) at IIASA. This collaborative effort ensured that the database was comprehensive and accurate. Subsequent updates were made in 2013 (HWSD v1.2) and in 2023 (HWSD v2.0).

HWSD v2.0 expands on its previous versions by including data from various national soil databases, offering detailed soil attributes for seven different soil layers (a significant increase from the two layers in HWSD v1.2), and adopting a consistent soil reference system combining FAO1990 standards with the World Reference Base for Soil Resources.

In September 2023, a revised version of HWSD v2.0 was released. For detailed information on the updates in this version (v2.01), refer to the technical report below.

The HWSD v2.0 features a GIS raster image file that is connected to a soil attribute database. This database provides comprehensive information on the composition of soil units in nearly 30,000 soil mapping units. The HWSD v2.0 Viewer, included with the database, automatically establishes this link, allowing easy access to both the soil attribute data and mapping information.

Download : [Download viewer & data \(only soil types\)](#) | [Download database \(.mdb\)](#) | [HWSD Raster v2.0](#) | [Technical Report and Instructions](#)

Changelog

1. Soil data from ISRIC - WISE replaced with WISE30sec database data, increasing soil profiles from 10,250 to 21,000.

Sindh Irrigation GIS Portal

← → ↺ irrigation.sindh.gov.pk/gis/gismaps ☆ 📄 ⌂ ⌚ 🗑️ 📱 🔍 🔄 | School

I have a strong fishy... GEE Tax ClimateData KARACHI: Karrar fa... AdvGIS Data HEC RS ACIAR-LEARN Databases Events WaterGovernance AI-Platforms MWF ideas >> All Bookmar

☰ Irrigation Department 🏠 Home ▼ Type here 🔍 32.3943 N , 62.6177 E 🔍 🔍 🔍 🔍

 **GIS Portal - Beta**

- ☑ Sindh
 - ▶ ☑ Irrigation System
 - ▶ ☑ Drainage System
 - ▶ ☑ Bunds
 - ▶ ☑ Small Dams
 - ▶ ☑ Lakes & Reservoir
 - ▶ ☑ Boundaries
 - ☑ Tubewells
 - ☑ Water Courses
 - ☑ Thar
- ▶ ☑ Rivers
 - ☑ Indus
 - ☑ Jhelum
 - ☑ Chenab
 - ☑ Ravi
 - ☑ Sutlej
 - ☑ Panjnad
 - ☑ TB Link
 - ☑ CJ Link
- ▶ ☑ Barrages
- ▶ ☑ Deserts

SIDA FO's Profile ▼ 📍

Satellite

Pakistan

<https://irrigation.sindh.gov.pk/gis>

27





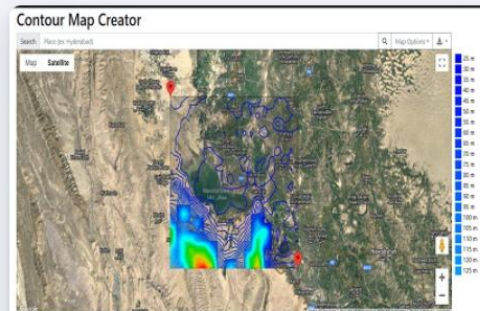
CLOUD APPS GALLERY

Experimental



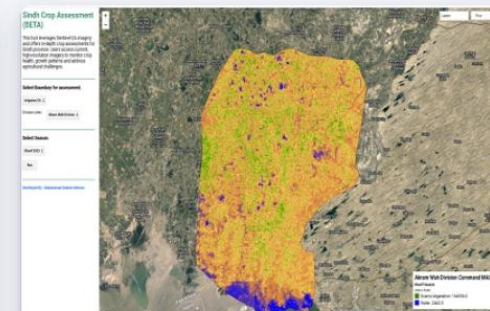
Irrigation GIS Portal

The GIS Portal offers comprehensive geospatial data, facilitating informed decision-making for efficient water resource management.



Contour Mapping

This app allows efficient visualization and analysis of land elevations, aiding in planning and water evacuation.



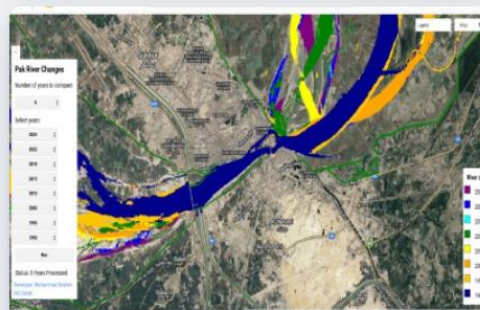
Sindh Crop Assessment

This tool leverages Sentinel-2A imagery and offers in-depth crop assessments for Sindh province.



Pak River Path

This tool leverages Sentinel-2A and Landsat imagery and offers in-depth Pak rivers .



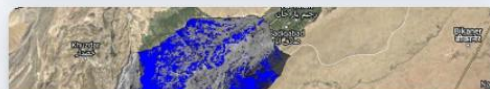
Pak River Path

This tool leverages Landsat imagery and offers changes in rivers path .



ACWA Portal Beta

GCF-FAO Project ACWA Portal combines field and remote sensing data.





Search for locations

Explore Data

Upload

Your workbench is empty

Click 'Explore Data' above to

- Browse the Data Catalogue
- Load your own data onto the map

TIP All your active data sets will be listed here

Analysis Story Map Share / Print English Help Login



Pakistan



New look



search in FAO Hand-in-Hand



Land
cover
(Khyber
Pakhtunk
hawa
Province,
Pakistan)



Land
cover
(Punjab
Province,
Pakistan)



Land
cover
(Sindh
Province,
Pakistan)



Pakistan

Description

(revised: 2026-07-14T17:30:35.119465)

- Built-up (BUP)
- Roads (ROA)
- Tree Orchards (ORC)
- Herbaceous crops irrigated (HCI)
- Herbaceous crops rainfed (HCR)
- Herbaceous crops in flood plain (HCF)
- Herbaceous natural vegetation (HER)
- Shrubs sparse natural vegetation (SOP)
- Shrubs dense natural vegetation (SCL)
- Shrubs in temporary wet soil (SWE)
- Trees sparse natural vegetation (TOP)
- Tree forest plantations (TFP)
- Trees dense natural vegetation (TCL)
- Sand dunes (SDU)
- Bare soil (BSO)
- Bare soil in temporary wet (BTW)
- Water bodies (WBO)
- Wetlands (WET)
- Snow (SNW)

Land cover (Provinces of Pakistan - 10m)

 Search for locations

 Explore Data  Upload

Map 

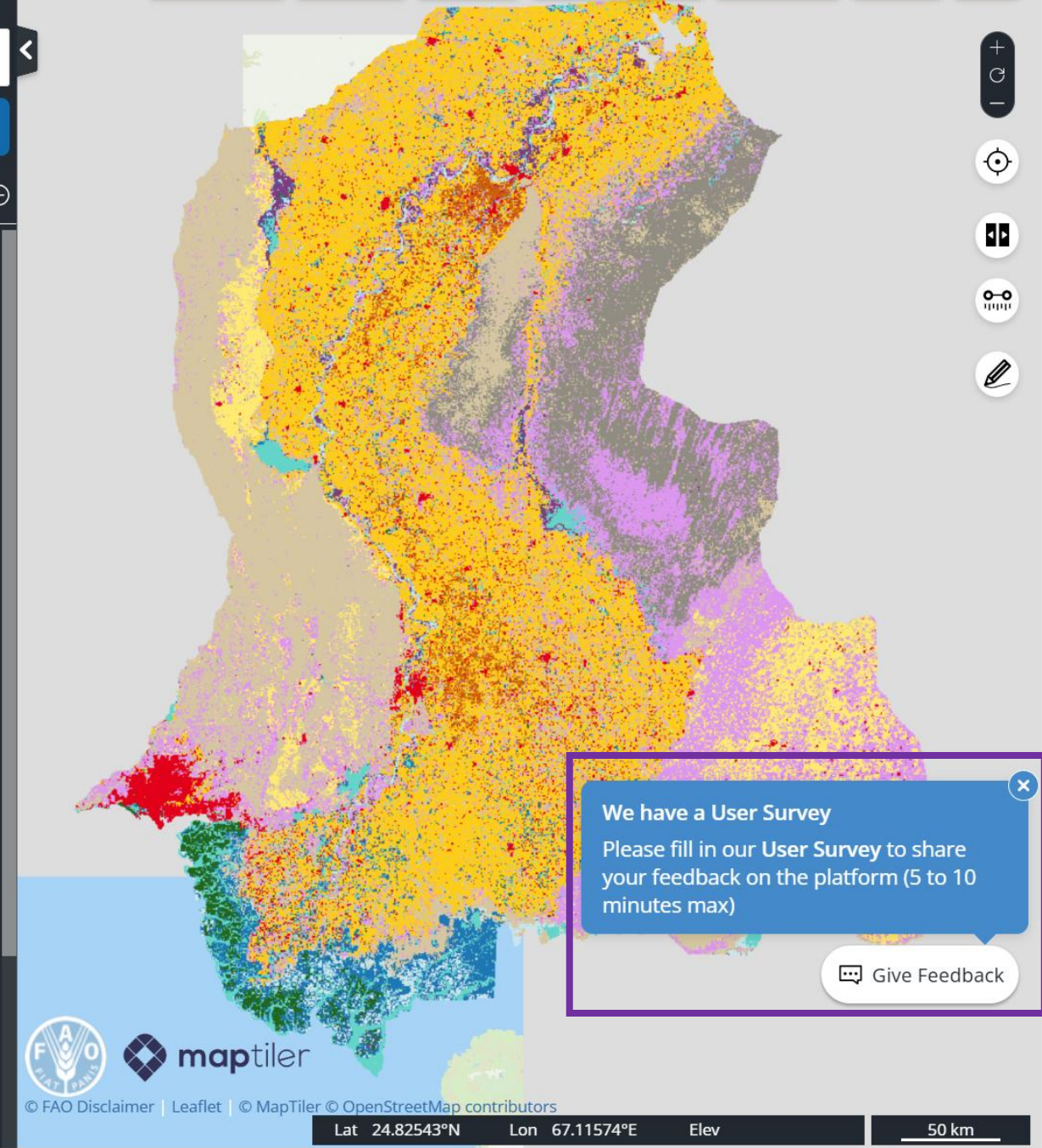
☒ Land cover (Sindh Province, Pakistan - 10 m) 

[Zoom To Extent](#) [About This Data](#) [Split](#) [Remove](#) 

Opacity: 100 % 

-  Built-up (BUP)
-  Roads (ROA)
-  Tree Orchards (ORC)
-  Herbaceous crops irrigated (HCI)
-  Herbaceous crops rainfed (HCR)
-  Herbaceous crops in flood plain (HCF)
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-  Shrubs dense natural vegetation (SCL)
-  Shrubs in temporary wet soil (SWE)
-  Trees sparse natural vegetation (TOP)
-  Tree forest plantations (TFP)
-  Trees dense natural vegetation (TCL)
-  Sand dunes (SDU)



Explore this site and fill out the survey form

We have a User Survey

Please fill in our **User Survey** to share your feedback on the platform (5 to 10 minutes max)



Other sources

The screenshot shows a web browser window with the URL <https://guides.libraries.psu.edu/c.php?g=376207&p=5296088>. The browser's address bar and tabs are visible at the top. The main content area features the Penn State University Libraries logo and a navigation menu with links to SERVICES, RESEARCH, ABOUT, and ASK. Below the navigation menu is a breadcrumb trail: Penn State University Libraries / Library Guides / Geography and Earth Sciences / Maps & Geospatial: Geographic Information Systems (GIS) / Finding GIS Data: International. The main heading is "Maps & Geospatial: Geographic Information Systems (GIS)", followed by the subtitle "Resources to discover GIS data, learn about GIS, and provide references for research." On the left side, there is a vertical menu with links to Introduction, Finding GIS Data: Pennsylvania, Finding GIS Data: US, Finding GIS Data: International, Library Resources, and ESRI Resources. The main content area is titled "Select Regional Sources" and lists several resources: Inspire Geoportal (European Union), EuroGeographics Open Data, European Union Open Data Portal, and European Environment Agency. Data and Maps. A small "AS A LIB" logo is visible on the right side of the page.

← → ↻ <https://guides.libraries.psu.edu/c.php?g=376207&p=5296088>

★ Bookmarks Quran English Trans... HP Connected Maayeka: Homema... Esri Technical Certifi... Arjumand Zaidi How to Make Chick... 17 Free GIS ESRI Co...

PennState
University Libraries

SERVICES RESEARCH ABOUT ASK

Penn State University Libraries / Library Guides / Geography and Earth Sciences / Maps & Geospatial: Geographic Information Systems (GIS) / Finding GIS Data: International

Maps & Geospatial: Geographic Information Systems (GIS)

Resources to discover GIS data, learn about GIS, and provide references for research.

Introduction

Finding GIS Data: Pennsylvania

Finding GIS Data: US

Finding GIS Data: International

Library Resources

ESRI Resources

Select Regional Sources

- [Inspire Geoportal \(European Union\)](#)
The INSPIRE geoportal enables search of spatial data and services, with subject to access restrictions.
- [EuroGeographics Open Data](#)
Open data available for Europe at 1: 1 million scale, including administrative boundaries, water networks, transportation, developments, and place names.
- [European Union Open Data Portal](#)
Open data is searchable by keyword and theme, covering environmental variables of interest or more.
- [European Environment Agency. Data and Maps](#)

AS
A LIB

Activate Windows

Data For Pakistan





marks



Quran English Trans...



HP Connected



Maayeka: Homema...



Esri Technical Certifi...



Arjumand Zaidi



How to Make Chick...



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Homework: Exploring Open RS and GIS Data

Objective: To identify, access, and evaluate freely available geospatial datasets for water resources applications.

Task: Search five websites that provide free RS, GIS, or hydro-meteorological datasets. Select at least one dataset from each website that could be useful for a Hydroinformatics application, such as rainfall analysis, flood mapping, drought monitoring, groundwater assessment, or irrigation management.

For details, please refer to the [Homework Workbook](#).



Discussion/Comments/
Questions?

